

Predicting Sleep Disturbance Risk from Gaming Behavior Using Machine Learning

Plaban Basu
BRAC University
ID: 22101831
plaban.basu@g.bracu.ac.bd

Minhaz Uddin
BRAC University
ID: 24241197
minhaz.uddin@g.bracu.ac.bd

Yeamin Alam
BRAC University
ID: 23241007
yeamin.alam@g.bracu.ac.bd

Abstract—Children and teenager are gradually spending a huge amount of time playing video games, which raises concerns about the impact of gaming on sleep health, specially among the university students. This study investigates the relationship between gaming behavior and sleep disturbance risk using a supervised machine learning approach applied to two independent dataset and a Bangladeshi university student gaming behavior survey. Three binary classification models- Support Vector Machine(SVM), XGBoost and Artificial Neural Network(ANN) are trained and assessed using using balanced accuracy, weighted F1-score and ROC-AUC. Results show that gaming-related behavioral features have strong predictive signal for sleep quality classification. Although model results and the best performing algorithm vary across different datasets as XGBoost is at the top, ANN is second. This shows the importance of dataset specific model selection in behavioral health prediction.

Index Terms—sleep disturbance, gaming behavior, machine learning, classification, support vector machine, XGBoost, artificial neural network, sleep quality

I. INTRODUCTION

The proliferation of video gaming as a primary leisure activity has raised growing concerns about its effects on sleep health and daily functioning. Research consistently shows that excessive gaming is associated with delayed sleep onset, shorter sleep duration, and increased nocturnal disturbances [1], [2]. These effects are particularly prominent among young adults and university students, who represent the largest demographic of active gamers worldwide.

Despite a substantial body of epidemiological evidence linking gaming behavior to sleep disorders, automated early-detection tools remain underdeveloped. Machine learning offers a promising path forward: by learning from behavioral, demographic, and lifestyle features, classifiers can flag individuals at elevated sleep-disturbance risk before symptoms become clinically severe [4], [7].

This paper frames sleep-disturbance prediction as a binary classification problem and benchmarks three algorithms - SVM, XGBoost, and ANN - across two demographically distinct datasets. Dataset 1 (Gaming & Mental Health, Kaggle, $n = 1,000$) captures global gaming and health indicators, with a binary target derived from self-reported sleep hours. Dataset 2 (Student Gaming Behavior Survey, Bangladesh, $n =$

989 after cleaning) provides survey data from Bangladeshi university students, enabling cross-population generalizability analysis with a target derived from self-reported bedtime.

The key contributions of this work are: (i) a reproducible end-to-end ML pipeline applied to both datasets, covering data loading, cleaning, encoding, SMOTE-based balancing, scaling, training, and evaluation; (ii) a systematic head-to-head comparison of SVM, XGBoost, and ANN on two demographically distinct data sources; and (iii) identification of the best-performing model for sleep disturbance classification from gaming behavioral data in each setting.

II. RELATED WORK

Research on gaming behavior and health outcomes spans clinical surveys, epidemiological studies, and ML-based prediction, providing strong motivation for this work.

Gaming and sleep health. A European survey of 109 adults aged 18–40 [1] applied the Gaming Addiction Scale (GAS) and Pittsburgh Sleep Quality Index (PSQI) and found that 25.7% qualified as addicted gamers, with 59.6% reporting poor sleep quality. Higher addiction scores directly predicted greater sleep disruption and increased reliance on sleep medication. A cross-sectional study from Pakistan [2] of 618 adults found that the 12.5% classified as gaming-addicted reported shorter sleep durations, more frequent nocturnal awakenings, and greater daytime dysfunction than non-addicted players.

ML for gaming-related health classification. Hasan et al. [3] collected data from 1,196 Bangladeshi children and applied KNN, SVM, Naive Bayes, and Random Forest to classify mental health impact levels from mobile gaming behavior. SVM achieved 92.75% accuracy, demonstrating that self-reported behavioral features can reliably support ML classifiers for gaming-related health outcomes, directly motivating our SVM baseline.

ML pipelines for behavioral health data. Syed et al. [4] reviewed how supervised ML models handle complex behavioral datasets including sedentary behavior and sleep patterns from wearable devices, outlining standard pipeline stages that inform our methodology. Van der Berg [5] used smartphone app-usage logs to predict sleep efficiency via

Logistic Regression, Random Forest, XGBoost, and MLP, confirming that daily gaming hours are a robust predictor feature and that tree-based ensembles compete favorably with neural networks on behavioral data.

Gaming intensity and wellbeing. Ahmed et al. [6] applied Logistic Regression, Random Forest, and K-Means Clustering to a university student dataset and found that gaming intensity and platform type significantly affect psychological wellbeing, validating `daily_gaming_hours` and `gaming_platform` as strong predictive features, both present in Dataset 1.

Multi-model sleep disorder classification. Rahman et al. [7] benchmarked Random Forest, XGBoost, SVM, and Decision Trees against lifestyle data to distinguish None, Insomnia, and Sleep Apnea categories, providing a direct architectural reference for our evaluation design. Islam et al. [8] applied an MLP to 500 PUBG players across gaming intensity levels and found that gaming duration significantly affects sleep quality, informing our ANN configuration.

Advanced sleep prediction techniques. Zhao et al. [9] developed an XGBoost model with SHAP feature-importance analysis achieving AUROC > 0.897 for predicting insomnia and sleep apnea from questionnaire data, demonstrating that survey features suffice for high-performance sleep-disorder prediction. Hossain et al. [10] compared four classifiers on a lifestyle dataset for sleep classification, underscoring that multi-model evaluation is essential — a principle central to this study.

III. DATASETS AND PREPROCESSING

A. Dataset 1 — Gaming and Mental Health

Source and structure. Dataset 1 was sourced from Kaggle [11] and contains 1,000 records with 27 features covering demographic information (age, gender), gaming behavior (daily gaming hours, game genre, gaming platform, primary game, years gaming, monthly spending), self-reported health indicators (sleep hours, sleep disruption frequency, eye strain, back/neck pain, weight change, exercise hours), and psychological markers (mood state, mood swing frequency, gaming addiction risk level, withdrawal symptoms, loss of other interests, continued despite problems, social isolation score, grades/GPA, work productivity score, academic/work performance).

Target variable. The binary classification target `sleep_target` was constructed from the `sleep_hours` column. A record was labeled *Good Sleep* (1) if the individual reported ≥ 7 hours of sleep per night, consistent with adult sleep health guidelines, and *Poor Sleep* (0) otherwise. This yielded 183 Good Sleep (18.3%) and 817 Poor Sleep (81.7%) instances, reflecting notable class imbalance. After constructing the target, the source columns `sleep_hours` and `sleep_quality` were both dropped to prevent data leakage during model training.

Column removal. Two non-predictive columns were removed: `record_id` (a unique row identifier with no predictive value) and `primary_game` (a high-cardinality free-

text game title that would introduce sparse one-hot encodings without meaningful signal).

Missing value imputation. Two columns contained missing values: `grades_gpa` (246 missing, 24.6%) and `work_productivity_score` (326 missing, 32.6%). All numerical missing values were filled with the column median; all categorical missing values were filled with the column mode.

Ordinal encoding. The following features were mapped to integer ordinal scales reflecting their natural ordering:

- `sleep_disruption_frequency`: Never=0, Rarely=1, Sometimes=2, Often=3, Always=4
- `academic_work_performance`: Failing/Poor=0 or 1, Average=2, Good=3, Excellent=4
- `mood_swing_frequency`: Never=0, Rarely=1, Sometimes=2, Often=3, Daily=4
- `gaming_addiction_risk_level`: Low=0, Moderate=1, High=2, Severe=3

Boolean indicator columns (`withdrawal_symptoms`, `loss_of_other_interests`, `continued_despite_problems`, `eye_strain`, `back_neck_pain`) were cast from boolean to integer (False=0, True=1).

One-hot encoding. The four nominal categorical columns — `gender`, `game_genre`, `gaming_platform`, and `mood_state` — were one-hot encoded using `drop_first=True` to avoid multicollinearity, yielding a final post-encoding feature space of 38 columns.

Noise injection and scaling. A small Gaussian noise term ($\mu = 0$, $\sigma = 0.1$) was added to `daily_gaming_hours` to reduce the risk of overfitting on an exactly repeated value. All features were then standardized to zero mean and unit variance using `StandardScaler` fitted on the training split only.

B. Dataset 2 — Student Gaming Behavior, Bangladesh

Source and structure. Dataset 2 [12] contains 1,062 raw survey responses from Bangladeshi university students collected in March–April 2023 across 20 features. The original survey column names were first renamed to clean Python identifiers: e.g., “When you go to sleep?” became `sleep_time`, “The average time you spend playing games?” became `avg_gaming_hours`, “How you fill when you can not play game in whole day?” became `feel_no_game`, and so on.

Outlier removal. Three sequential filtering steps were applied before any other preprocessing:

- 1) 22 rows with CGPA outside $[0, 4]$ or HSC result outside $[0, 5]$ were removed as data-entry errors (e.g., HSC results up to 4382 were present in the raw data).
- 2) 26 rows with `social_hours` > 7 were dropped as implausible values (likely entry errors).
- 3) The `addicted_game` free-text column was parsed against a seven-item whitelist (PUBG, FREE FIRE, CALL OF DUTY, IGA, PES, FIFA, COC) to create a clean `primary_game` categorical column. 25

rows containing unrecognized game names were removed. The original `addicted_game` column was then dropped.

After these steps, 989 records remained.

Bedtime parsing. The `sleep_time` column stored responses as AM/PM time strings (e.g., “2:00:00 AM”, “11:00:00 PM”). Each value was parsed into a 24-hour integer bedtime hour using `pd.to_datetime` with AM/PM format inference. An additional correction was applied: hours reported between 07:00 and 11:00 were shifted by +12 to map them from apparent AM to their PM equivalent (correcting survey ambiguity), and 12:00 was remapped to 0. Rows where the bedtime could not be parsed were dropped, leaving 989 usable records.

Target variable. The binary target `sleep_target` was derived from the parsed bedtime hour. A student was labeled *Good Sleep* (1) if their bedtime fell at or before 01:00 (i.e., $\text{hour} \geq 20$ or $\text{hour} \leq 1$), and *Poor Sleep* (0) for all other hours (sleeping after 2 AM or at unusual daytime hours). This yielded 785 Good Sleep (79.4%) and 204 Poor Sleep (20.6%) instances. The `sleep_hour` column was subsequently dropped.

Emotional response encoding. The column `feel_no_game` (“How you feel when you cannot play games in a whole day?”) was mapped to a three-level sentiment scale: Positive responses (“Positive”, “Alright”, “Not Bad”, “Calm”) = 2; Negative responses (“Upset”, “Depressed”, “Angry”, “Bore”, “Bored”) = 0; all others = 1.

Ordinal encoding. The following binary and ordinal columns were label-encoded:

- `feel_complete`: Bored=0, Excited=1
- `feel_incomplete`: Stressed=0, No feeling=1, Excited to complete level=2
- `morning_class`, `reads_newspaper`, `fatigue`, `stress_relief`, `wears_glasses`: Yes/Regular=1, No/Not regular=0
- `game_type`: Non Paid=0, Paid=1

One-hot encoding. The four nominal columns — `edu_level`, `gender`, `platform`, and `primary_game` — were one-hot encoded with `drop_first=True`, yielding a final feature space of 25 columns.

Missing value imputation and scaling. Any remaining numerical missing values after encoding were filled with column medians. All features were then standardized using `StandardScaler` fitted on the training split only.

C. Train-Test Split and Class Balancing

Both datasets were split 80/20 using stratified sampling (`random_state = 42`) to preserve class proportions. Dataset 1 yielded 800 training and 200 test samples; Dataset 2 yielded 791 training and 198 test samples. SMOTE (Synthetic Minority Oversampling Technique) [7] was applied exclusively to the training split after scaling to generate synthetic minority-class samples, preventing any leakage of synthetic data into the held-out test set.

IV. METHODOLOGY

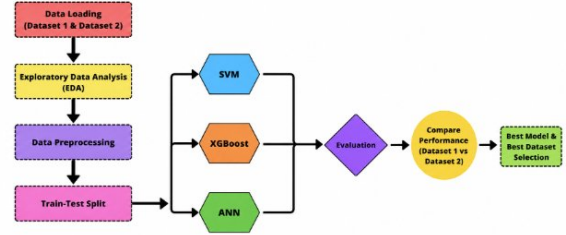


Fig. 1. End-to-end ML pipeline applied to both datasets.

A. Overall Pipeline

The end-to-end pipeline follows six sequential stages applied identically to both datasets: (1) Data Loading and column renaming, (2) Exploratory Data Analysis (EDA) including distribution plots, correlation heatmaps, and boxplots, (3) Preprocessing as described in Section III, (4) stratified Train-Test Split with SMOTE balancing, (5) model training and evaluation of SVM, XGBoost, and ANN, and (6) cross-dataset comparison to select the best model per dataset.

B. Support Vector Machine

A Support Vector Classifier with a radial basis function kernel (`kernel='rbf'`, `random_state=42`) was trained on the SMOTE-balanced training data. For Dataset 2, `probability=True` was additionally set to enable ROC-AUC computation via `predict_proba`; for Dataset 1 the SVM was trained without probability calibration and the decision function was used for AUC scoring instead. The RBF kernel implicitly maps features into a high-dimensional space, enabling non-linear decision boundaries appropriate for the mixed ordinal and one-hot encoded feature spaces in both datasets.

C. XGBoost

An extreme gradient-boosted tree ensemble was trained with dataset-specific hyperparameters. For Dataset 1: `learning_rate=0.01`, `n_estimators=1000`, `max_depth=4`, `subsample=0.8`, `colsample_bytree=0.3`, `reg_alpha=10`, `reg_lambda=5`, `eval_metric=logloss`. For Dataset 2: `learning_rate=0.1`, `n_estimators=300`, `max_depth=5`, `subsample=0.9`, `colsample_bytree=0.8`, `reg_alpha=0.1`, `reg_lambda=0.5`, `eval_metric=logloss`. Dataset 1 required a slower learning rate and much stronger L1 regularization (`reg_alpha=10` vs 0.1) to prevent overfitting on its larger post-SMOTE and higher-dimensional feature space (38 features), while Dataset 2 was trained with a faster learning rate and lighter regularization given its smaller feature space (25 features) and lower imbalance ratio.

D. Artificial Neural Network

A Multilayer Perceptron (MLP) with two hidden layers of sizes (64, 32) neurons was trained using scikit-learn’s `MLPClassifier` with `max_iter=1000` and `random_state=42`. Default settings were used for all other hyperparameters: ReLU activation function, Adam optimizer with an initial learning rate of 0.001, L_2 regularization parameter $\alpha = 0.0001$, and a mini-batch size of 200. The same architecture and hyperparameters were applied identically to both datasets. The network learns hierarchical non-linear representations of the input features through two fully connected layers before the final sigmoid output layer for binary classification.

E. Evaluation Metrics

Each model was evaluated on the held-out test set using five metrics. **Balanced Accuracy** (primary metric) is the arithmetic mean of per-class recall and is robust to class imbalance, penalizing models that ignore the minority class. **Accuracy** measures the fraction of all correctly classified samples. **Precision** for the positive class measures the fraction of predicted positives that are true positives. **Recall** measures the fraction of actual positives correctly identified. **F1-Score** is the harmonic mean of precision and recall. **ROC-AUC** measures the area under the receiver operating characteristic curve across all classification thresholds. Confusion matrices were additionally produced for all six model-dataset combinations.

V. RESULTS

A. Dataset 1 — Gaming and Mental Health

Table I reports the overall performance of all three classifiers on the 200-sample test set of Dataset 1. The class distribution in the test set was 163 Poor Sleep and 37 Good Sleep, preserving the 81.7%/18.3% imbalance of the full dataset.

TABLE I
MODEL PERFORMANCE — DATASET 1 (GAMING & MENTAL HEALTH)

Model	Bal. Acc	Acc	Wtd. F1	ROC-AUC
SVM	0.9337	0.9600	0.9600	0.9801
XGBoost	0.9742	0.9750	0.9800	0.9985
ANN	0.9140	0.9450	0.9500	0.9796

Bold = best per metric. Test set: 200 samples.

Table II presents per-class precision, recall, and F1-score for Dataset 1.

XGBoost achieved the strongest performance on Dataset 1 across all reported metrics, attaining a balanced accuracy of 0.9742, a recall of 0.97 for the minority *Good Sleep* class, and a near-perfect ROC-AUC of 0.9985. SVM performed competitively with a balanced accuracy of 0.9337 and ROC-AUC of 0.9801. ANN followed with a balanced accuracy of 0.9140 and ROC-AUC of 0.9796.

TABLE II
PER-CLASS CLASSIFICATION REPORT — DATASET 1

Model	Class	Prec.	Rec.	F1	Supp.
SVM	Poor Sleep	0.98	0.98	0.98	163
	Good Sleep	0.89	0.89	0.89	37
XGBoost	Poor Sleep	0.99	0.98	0.98	163
	Good Sleep	0.90	0.97	0.94	37
ANN	Poor Sleep	0.97	0.96	0.97	163
	Good Sleep	0.84	0.86	0.85	37

B. Dataset 2 — Student Gaming Behavior, Bangladesh

Table III presents results on the 198-sample test set of Dataset 2. The test distribution was 157 Good Sleep (early bedtime) and 41 Poor Sleep (late bedtime), preserving the 79.4%/20.6% split.

TABLE III
MODEL PERFORMANCE — DATASET 2 (IGD SURVEY, BANGLADESH)

Model	Bal. Acc	Acc	Wtd. F1	ROC-AUC
SVM	0.8271	0.8687	0.8700	0.9264
XGBoost	0.8552	0.8990	0.9000	0.9492
ANN	0.9104	0.9293	0.9300	0.9414

Bold = best per metric. Test set: 198 samples.

Table IV presents per-class metrics for Dataset 2.

TABLE IV
PER-CLASS CLASSIFICATION REPORT — DATASET 2

Model	Class	Prec.	Rec.	F1	Supp.
SVM	Poor Sleep (late)	0.66	0.76	0.70	41
	Good Sleep (early)	0.93	0.90	0.92	157
XGBoost	Poor Sleep (late)	0.74	0.78	0.76	41
	Good Sleep (early)	0.94	0.93	0.94	157
ANN	Poor Sleep (late)	0.80	0.88	0.84	41
	Good Sleep (early)	0.97	0.94	0.95	157

ANN achieved the highest balanced accuracy (0.9104) and overall accuracy (92.93%) on Dataset 2, with XGBoost leading on ROC-AUC (0.9492). SVM showed the lowest balanced accuracy (0.8271), with the minority *Poor Sleep* class achieving only F1 = 0.70.

C. Cross-Dataset Comparison

Table V aggregates all six model-dataset results, enabling direct comparison of the same algorithm across the two datasets.

Fig. 2 visualises the four key metrics side-by-side for both datasets. Comparing each model across the two datasets, SVM drops by 10.66 percentage points in balanced accuracy (0.9337 on DS1 vs. 0.8271 on DS2), XGBoost drops by 11.90 points (0.9742 vs. 0.8552), while ANN drops by only 0.36

TABLE V
FULL CROSS-DATASET MODEL COMPARISON

DS	Model	Bal. Acc	Acc	Prec	Rec	AUC
DS1	SVM	0.9337	0.9600	0.8919	0.8919	0.9801
DS1	XGBoost	0.9742	0.9750	0.9000	0.9730	0.9985
DS1	ANN	0.9140	0.9450	0.8421	0.8649	0.9796
DS2	SVM	0.8271	0.8687	0.9338	0.8981	0.9264
DS2	XGBoost	0.8552	0.8990	0.9419	0.9299	0.9492
DS2	ANN	0.9104	0.9293	0.9673	0.9427	0.9414

DS1 test: $n = 200$; DS2 test: $n = 198$.

Prec/Rec are for the positive (Good Sleep) class.

points (0.9140 vs. 0.9104) — making ANN by far the most consistent model across both datasets. Notably, the model ranking reverses entirely between the two datasets: XGBoost > ANN > SVM on Dataset 1, but ANN > XGBoost > SVM on Dataset 2.

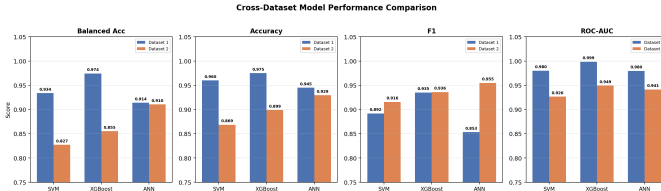


Fig. 2. Cross-dataset comparison of all three models across Balanced Accuracy, Accuracy, F1-Score, and ROC-AUC. Blue = Dataset 1, Orange = Dataset 2.

VI. DISCUSSION

Why XGBoost dominates Dataset 1. Dataset 1 contains features that are structurally close to the target label, particularly `sleep_disruption_frequency` (an ordinal scale of how often the user is woken at night) and `gaming_addiction_risk_level` (a composite risk score). These two features provide an almost direct path to predicting whether a person sleeps fewer than 7 hours. XGBoost’s boosted tree ensemble is highly effective at exploiting a small number of dominant features across a large post-SMOTE training set (38 features, 1,000 records), and its strong L1 regularization (`reg_alpha=10`) prevents the remaining weaker features from introducing noise. This explains why it achieves a recall of 0.97 for the minority Good Sleep class and a ROC-AUC of 0.9985.

Why ANN leads on Dataset 2. Dataset 2 targets are derived from self-reported bedtime rather than sleep duration, and the available features are entirely indirect — CGPA, gaming start age, social hours, emotional responses when unable to play, and platform type. No feature directly measures sleep disturbance. In this more ambiguous, lower-signal setting, the MLP’s two hidden layers (64 and 32 neurons) allow it to learn complex non-linear combinations of these weak signals that neither a linear SVM kernel nor a shallow boosted tree can capture as effectively. This pattern is consistent with Islam

et al. [8], who found that neural networks generalize better than tree-based models when the connection between gaming behavior and health outcomes is indirect and mediated through multiple intermediate variables.

Why ANN is the most consistent model overall. While XGBoost and SVM both suffer significant balanced accuracy drops between Dataset 1 and Dataset 2 (11.90 and 10.66 percentage points respectively), ANN’s balanced accuracy changes by only 0.36 points (0.9140 on DS1 to 0.9104 on DS2). This stability is attributable to the MLP’s architecture: gradient descent over a smooth loss surface with L2 regularization avoids the extreme overfitting to dominant features that benefits XGBoost on DS1 but causes it to underperform on DS2’s weaker feature-target relationships.

Why SVM underperforms on Dataset 2. SVM achieves a minority-class F1 of only 0.70 on Dataset 2. The RBF kernel relies on a well-defined margin between classes, but Dataset 2’s mixed Boolean, ordinal, and one-hot encoded features create a high-dimensional, noisy feature space where the margin is diffuse. Without probability calibration or careful kernel hyperparameter tuning over C and γ , the default RBF-SVM cannot find a consistent boundary for the small minority class (41 late sleepers out of 198 test samples).

Practical implications. Three actionable insights emerge from these findings. First, when survey data includes explicit sleep-related health indicators such as sleep disruption frequency or addiction risk levels, XGBoost is the recommended classifier due to its ability to exploit dominant features efficiently. Second, when only indirect behavioral proxies (gaming habits, academic performance, emotional responses) are available, a two-layer MLP offers both the best performance and the greatest consistency across populations. Third, the reversal in model ranking between the two datasets underscores that no single model generalizes universally across different survey contexts, and dataset-specific validation is essential before deployment.

Limitations. XGBoost’s near-perfect performance on Dataset 1 warrants careful interpretation. Features such as `sleep_disruption_frequency` are conceptually very close to the target label (sleep quality), and their inclusion may constitute a form of label leakage. An ablation study removing these features is recommended. Both datasets rely entirely on self-reported data, introducing potential recall and social desirability biases. The binary target collapses nuanced sleep patterns into two categories; a multi-class formulation would yield finer-grained clinical utility. Finally, SMOTE-generated synthetic samples may not fully represent the true distribution of minority-class individuals.

VII. CONCLUSION

This paper presented a supervised machine learning study predicting sleep disturbance risk from gaming behavioral data across two independent datasets using a standardized six-stage pipeline. On Dataset 1, XGBoost was the strongest model (balanced accuracy 0.9742, ROC-AUC 0.9985), driven by the availability of direct sleep-health indicators in the

feature set. On Dataset 2, where only indirect behavioral proxies were available, ANN led (balanced accuracy 0.9104, accuracy 92.93%), with XGBoost achieving the highest ROC-AUC (0.9492). ANN proved the most consistent model overall, with a balanced accuracy drop of only 0.36 percentage points across the two datasets, compared to drops of 11.90 and 10.66 percentage points for XGBoost and SVM respectively.

These findings confirm that gaming behavior features carry meaningful predictive signal for sleep quality classification, and that model selection must be guided by the nature and directness of the available features rather than by a one-size-fits-all choice. Future directions include SHAP-based feature importance analysis, ablation studies to quantify label leakage risk from sleep-adjacent features, multi-class sleep-stage classification, and cross-population transfer learning between the two datasets.

REFERENCES

- [1] T. Türkçapar, "The relationship between sleep problems and gaming addiction in adults," *European Psychiatry*, vol. 65, no. S1, 2022. <https://doi.org/10.1192/j.eurpsy.2022.2113>
- [2] A. Hussain et al., "Prevalence of gaming addiction and its impact on sleep quality: A cross-sectional study from Pakistan," 2022. <https://pubmed.ncbi.nlm.nih.gov/35734653>
- [3] T. Hasan et al., "Prediction of Bangladeshi urban children's mental health for the effect of mobile gaming using machine learning," *ResearchGate*, 2024. <https://www.researchgate.net/publication/381654411>
- [4] A. Syed et al., "Machine learning in physical activity, sedentary, and sleep behavior research," *J. Activity, Sedentary and Sleep Behaviors*, 2024. <https://doi.org/10.1186/s44167-024-00045-9>
- [5] R. van der Berg, "Predicting sleep quality from smartphone app usage using machine learning," Tilburg University, 2022. <http://arno.uvt.nl/show.cgi?fid=170844>
- [6] Author(s), "Impact of gaming habits on psychological wellbeing and academic risk in university students," *IEEE Xplore*, 2025. <https://ieeexplore.ieee.org/document/11380705/>
- [7] M. Rahman et al., "Evaluating machine learning models for predicting sleep disorders in a lifestyle and health data context," *ResearchGate*, 2024. <https://www.researchgate.net/publication/380275509>
- [8] M. Islam et al., "Gaming behavior and health outcomes in PUBG players: A machine learning approach," *BMC Public Health*, 2025. <https://doi.org/10.1186/s12889-025-21880-8>
- [9] Y. Zhao et al., "Machine learning-based prediction of sleep disorder risk using XGBoost and SHAP analysis," *PMC*, 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10557018/>
- [10] M. Hossain et al., "Evaluating machine learning models for predicting sleep disorders in a lifestyle and health data context," *ResearchGate*, 2024. <https://www.researchgate.net/publication/380275509>
- [11] S. Shahid, "Gaming and mental health dataset," Kaggle, 2024. <https://www.kaggle.com/datasets/shaistashahid/gaming-and-mental-health>
- [12] Author(s), "A multidimensional dataset of student gaming behavior and academic correlates in Bangladesh," *Scientific Data*, 2025. <https://doi.org/10.1038/s41597-025-05614-4>